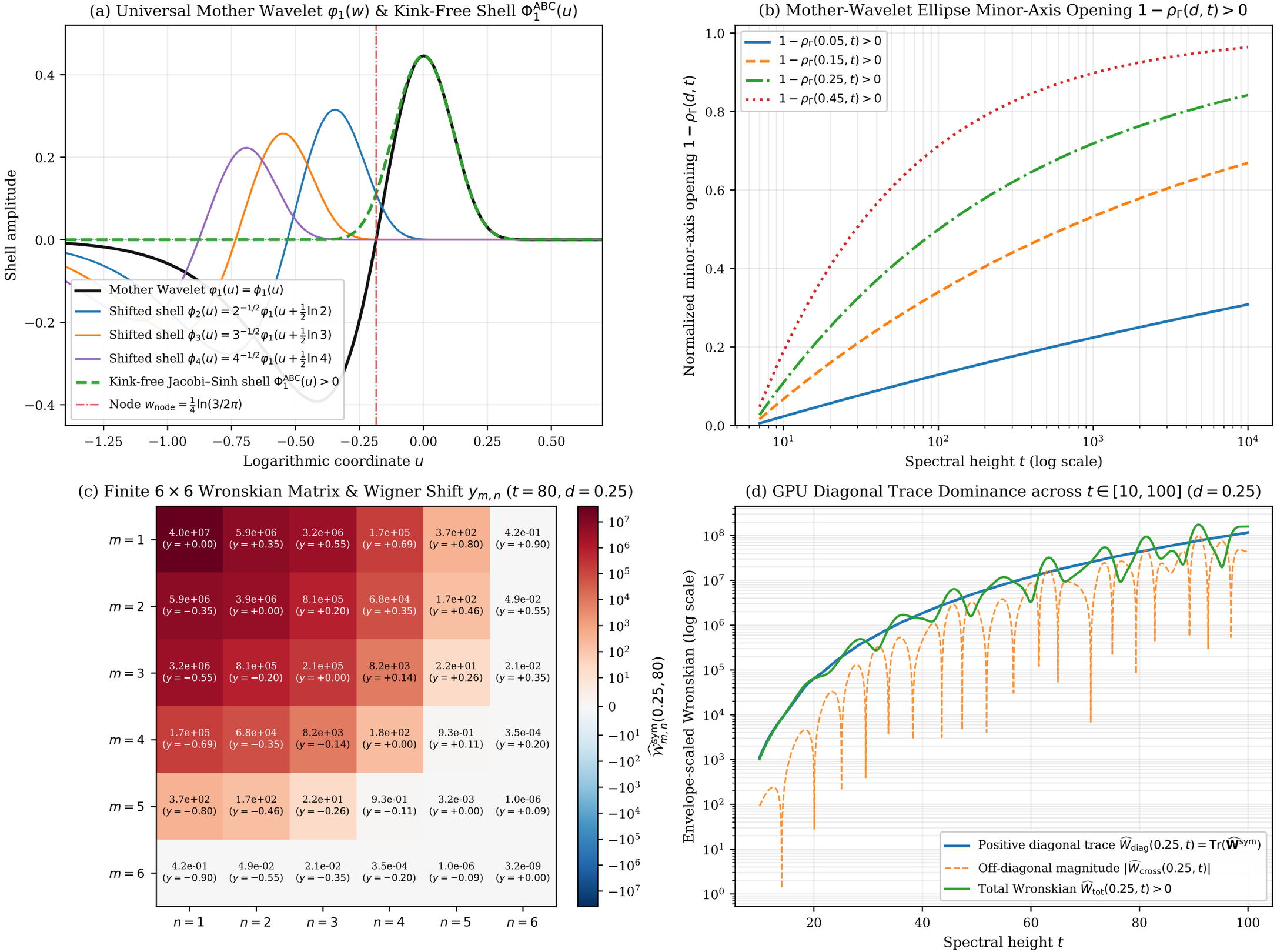

(a) Universal Mother Wavelet $\varphi_1(w)$ & Kink-Free Shell $\Phi_1^{\mathrm{ABC}}(u)$
Shell amplitude
Logarithmic coordinate u
Mother Wavelet $\varphi_1(u) = \phi_1(u)$
Shifted shell $\phi_2(u) = 2^{-1/2}\varphi_1(u + \frac{1}{2}\ln 2)$
Shifted shell $\phi_3(u) = 3^{-1/2}\varphi_1(u + \frac{1}{2}\ln 3)$
Shifted shell $\phi_4(u) = 4^{-1/2}\varphi_1(u + \frac{1}{2}\ln 4)$
Kink-free Jacobi–Sinh shell $\Phi_1^{\mathrm{ABC}}(u) > 0$
Node $w_{\mathrm{node}} = \frac{1}{4}\ln(3/2\pi)$
(b) Mother-Wavelet Ellipse Minor-Axis Opening $1 - \rho_\Gamma(d, t) > 0$
Normalized minor-axis opening $1 - \rho_\Gamma(d, t)$
Spectral height t (log scale)
$1 - \rho_\Gamma(0.05, t) > 0$
$1 - \rho_\Gamma(0.15, t) > 0$
$1 - \rho_\Gamma(0.25, t) > 0$
$1 - \rho_\Gamma(0.45, t) > 0$
(c) Finite 6×6 Wronskian Matrix & Wigner Shift $y_{m,n}$ $(t = 80, d = 0.25)$
$m = 1$ 4.0e+07 ($y = +0.00$) 5.9e+06 ($y = +0.35$) 3.2e+06 ($y = +0.55$) 1.7e+05 ($y = +0.69$) 3.7e+02 ($y = +0.80$) 4.2e-01 ($y = +0.90$)
$m = 2$ 5.9e+06 ($y = -0.35$) 3.9e+06 ($y = +0.00$) 8.1e+05 ($y = +0.20$) 6.8e+04 ($y = +0.35$) 1.7e+02 ($y = +0.46$) 4.9e-02 ($y = +0.55$)
$m = 3$ 3.2e+06 ($y = -0.55$) 8.1e+05 ($y = -0.20$) 2.1e+05 ($y = +0.00$) 8.2e+03 ($y = +0.14$) 2.2e+01 ($y = +0.26$) 2.1e-02 ($y = +0.35$)
$m = 4$ 1.7e+05 ($y = -0.69$) 6.8e+04 ($y = -0.35$) 8.2e+03 ($y = -0.14$) 1.8e+02 ($y = +0.00$) 9.3e-01 ($y = +0.11$) 3.5e-04 ($y = +0.20$)
$m = 5$ 3.7e+02 ($y = -0.80$) 1.7e+02 ($y = -0.46$) 2.2e+01 ($y = -0.26$) 9.3e-01 ($y = -0.11$) 3.2e-03 ($y = +0.00$) 1.0e-06 ($y = +0.09$)
$m = 6$ 4.2e-01 ($y = -0.90$) 4.9e-02 ($y = -0.55$) 2.1e-02 ($y = -0.35$) 3.5e-04 ($y = -0.20$) 1.0e-06 ($y = -0.09$) 3.2e-09 ($y = +0.00$)
$n = 1$ $n = 2$ $n = 3$ $n = 4$ $n = 5$ $n = 6$
$\widehat{\mathcal{W}}^{\mathrm{sym}}_{m,n}(0.25, 80)$
(d) GPU Diagonal Trace Dominance across $t \in [10, 100]$ $(d = 0.25)$
Envelope-scaled Wronskian (log scale)
Spectral height t
Positive diagonal trace $\widehat{W}_{\mathrm{diag}}(0.25, t) = \mathrm{Tr}(\widehat{\mathbf{W}}^{\mathrm{sym}})$
Off-diagonal magnitude $|\widehat{W}_{\mathrm{cross}}(0.25, t)|$
Total Wronskian $\widehat{W}_{\mathrm{tot}}(0.25, t) > 0$